



DIGITAL WEIGHT MEASURING MACHINE

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INTRODUCTION

This project demonstrates how to build a digital weight measuring machine using an Arduino, a load cell, and an HX711 amplifier. The system converts physical force into accurate digital readings, offering a simple and affordable way to understand how modern electronic weighing scales work. It's a practical hands-on example of sensor interfacing, signal processing, and microcontroller programming.

MAIN COMPONENTS

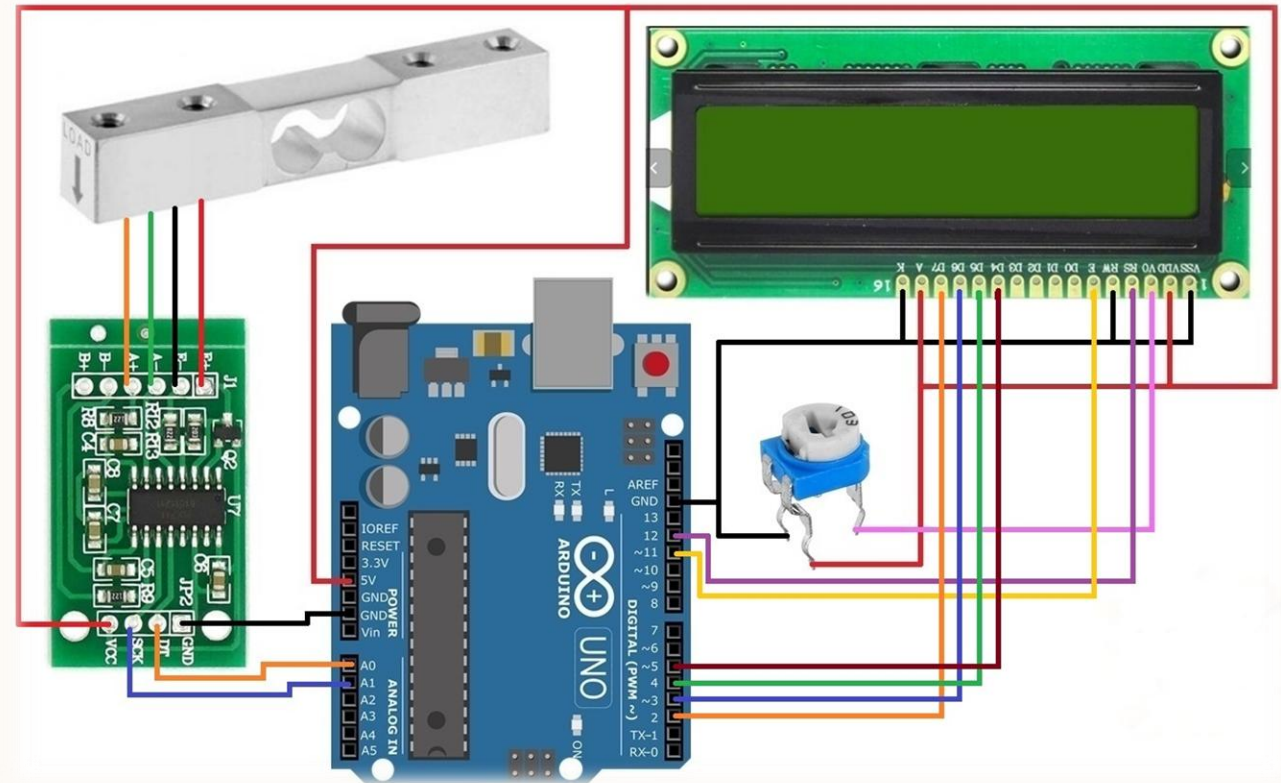
- **Arduino Uno** – the brain of the device
- **Load Cell Sensor** – detects force/weight
- **HX711 Amplifier Module** – converts load cell signal to readable data
- **Display (LCD or Serial Monitor)** – shows weight
- **Breadboard & Jumper Wires** – connects everything

HOW IT WORKS

When weight is placed on the load cell, it creates a tiny change in electrical resistance. The HX711 amplifier boosts this signal so the Arduino can read it. The Arduino then converts the signal into a weight value using a calibration factor and displays the result on the screen.

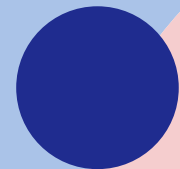
CIRCUIT DIAGRAM

The load cell outputs a tiny analog signal → HX711 amplifies it → Arduino reads digital data and sends output to the display



SUMMARY

This project uses an Arduino, a load cell, and an HX711 module to create a simple digital weighing machine. The load cell senses the applied weight, the amplifier prepares the signal, and the Arduino processes and displays the measurement. It's an easy, low-cost way to learn how electronic scales work while practicing sensor interfacing and basic programming.



**THANK
YOU**